4-Year Strategic Research Roadmap: Real-Time Aerodynamic State Estimation Using Surface Sensors and Data Assimilation

Overall Vision

Develop and experimentally validate a real-time data assimilation framework capable of predicting separated and dynamically stalled aerodynamic flows using only surface pressure/temperature sensors. The project culminates in a high-speed wind tunnel demonstration—directly relevant to aerospace applications such as maneuvering aircraft, aeroelastic stability, and flow control.

1. PHASE 1 (Year 1): Foundational Development + Canonical Validation

Goals

- Build the complete data assimilation (DA) pipeline end-to-end.
- Validate the approach on controlled, canonical flows before moving to complex airfoil dynamics.

Key Actions

1. Develop Minimal Viable DA Framework
   - Implement fast-enough ROM + EnKF/UKF/hybrid filter.
   - Demonstrate real-time update capability using synthetic data.
2. Experimental Validation on Cylinder Flow
   - Use surface pressure/temperature sensors to reconstruct vortex shedding dynamics.
   - Assess latency, robustness, noise handling.
   - Benchmark predictions vs. reference (PIV or CFD).

Deliverables

- Working DA pipeline (ROM + filter + real-time interface).
- Journal/Conference Paper: "Real-time estimation of vortex shedding dynamics from surface sensors using data assimilation."

2. PHASE 2 (Year 2): Main Contribution — Airfoil With Dynamic Stall

Goals

- Demonstrate robust real-time prediction of a dynamically changing aerodynamic state (stall onset, separation, reattachment, force coefficients).
- Produce the core results of the PhD.

Key Actions

1. Design and Execute Dynamic Stall Experiments.
   - Airfoil with controlled pitch ramps or oscillations.
   - Distributed surface pressure/temperature sensors.
   - Measurements of forces, possibly PIV/Schlieren for validation.
2. Real-Time State Estimation
   - Predict:
     - lift and moment in real time
     - flow regime (attached → pre-stall → stall → reattachment)
     - approximate separation location
   - Evaluate latency vs. characteristic flow timescales.
3. Model Refinement
   - Improve ROM structure (SPOD/DMD-based, nonlinear ROMs, hybrid ML-physics if beneficial).
   - Enhance DA stability under strongly nonlinear transitions.

Deliverables

- Core PhD dataset and results (airfoil dynamic stall).
- High-quality journal paper.
- 80–90 % of thesis chapters drafted by end of year.

3. PHASE 3 (Year 3): Transition to High-Speed Wind Tunnel Demonstration

Goals

- Adapt the method for compressible/high-Re conditions.
- Conduct first campaigns in the high-speed tunnel using simplified conditions.

Key Actions

1. Method Adaptation
   - Incorporate compressibility effects into the ROM, if necessary.
   - Sensor calibration for high-speed environment.
   - Robustness testing under increased noise and harsh conditions.
2. Experimental Campaign 1 in High-Speed Tunnel
   - Reproduce a simplified separated flow configuration (steady or mildly dynamic).
   - Validate real-time state estimation with surface sensors only.

Deliverables

- High-impact journal paper (AIAA Journal / Journal of Fluids Engineering).
- Proof-of-concept demonstration in relevant aerospace conditions.

4. PHASE 4 (Year 4): Final Demonstration + High-Speed Dynamic Stall / Buffet

Goals

- Execute an ambitious high-speed experiment showcasing the full capabilities of the method.
- Deliver the "flagshipresult connecting DA + real-time + separated compressible flow.

Key Actions

1. Full-Scale Demonstration
   - Airfoil with dynamic stall or moderate transonic buffet.
   - Real-time prediction of aerodynamic forces and flow regime transitions.
   - Demonstrate applicability to aerospace-relevant unsteady phenomena.
2. Final Synthesis
   - Integrate all components into a generalizable DA framework.
   - Produce final thesis and final set of publications.

Deliverables

- Final PhD thesis.
- "Capstone" publication: "Real-time aerodynamic state estimation in high-speed separated flows using surface sensors and data assimilation."
- Dataset + software framework for future research.

5. Expected Impact

Scientific

- Advances experimental data assimilation for highly nonlinear, unsteady aerodynamic flows.
- Provides a framework for real-time flow state reconstruction using minimal sensing.

Technological / Aerospace

Enables applications in:

- maneuver load prediction
- stall/buffet detection
- aeroelastic monitoring
- closed-loop flow control.

Strategic

Establishes the group as a leader in real-time experimental DA for aerodynamic systems.